

SYSTEMS THINKING

Systems Analysis & Design

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Outline

- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification



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Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.
- Not all elements need to be connected to each other, but every **connection** should be meaningful.
- The more **connections** there are, the more complex the system becomes. The representation must be feasible.
- Each element must have at least one connection. Isolated elements make no sense in a system.



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Complexity in Systems

System complexity can be defined as the **number of elements** and **connections** in a system.



Introduction to Systems Thinking II

- In **systems thinking**, if you just **split** the parts and forget the relationships, you will **lose the full picture**.
- It is called a **holistic** approach; try to see **the whole picture** with all the **meaningful details**.
- Defining the **box boundaries** is sometimes tricky; as we said, not too complex, not too simple. It is like the desired universe **balance** of Thanos.



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Introduction to Systems Thinking III

- Another important concept is **homeostasis**; it means putting a system in an **equilibrium state**. That is hard because **systems** are both **not in equilibrium** and **resilient to change**.
- A **system** is more than the **sum of its parts**. This means relationships, behaviors, and recovery capacity are **forgotten** when you see the system just as **its parts**.



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Introduction to Systems Thinking IV

- **Systems thinking** is a way to **understand** and **represent** problems in order to find the best possible **solution**.
- Thinking of a **problem as a system** lets you understand **details**, involved **elements**, and relevant **information**.
- **Systems** should be **viable**, **self-sustainable**, provide internal **feedback loops**, and also look like a **whole living entity**.



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Introduction to Systems Thinking V

- **Computation** helps to **represent behaviors** mathematically. Also, it helps to **find patterns** and information, and simplify processes; an example of all this is **Artificial Intelligence**.
- **Top-Down** approach is useful when you want to **see the full picture**, and then split it into parts.
- The **Bottom-Up** approach is useful when you want to **see the parts** and then connect them to get the full picture.



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Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to better decision-making.
- The simplest **system definition** is: given some **inputs**, after applying a designed **process** to them, you will obtain some **outputs**.
- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, real life is not deterministic.



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Introduction to Systems Thinking VII

- Since **randomness** is normal in the real world, relying solely on deterministic processes is **dangerous**. Using **stochastic processes** is a better approach.
- **Stochastic processes** make use of **probability**, which leads to a better representation of real-world behavior.
- Here, **Chaos Theory** becomes a useful **tool**. To put it simply, chaos can be defined as a harmonious **balance** between **rules** and **randomness**.



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Systems Structure Draw



Case of Study: Transportation System



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Systems Properties I

- **Emergence** is a property of systems that means that the **whole system** is more than the **sum of its parts**.
- **Interconnectedness** is a property of systems that means that all the **elements** are **connected** in a **meaningful** way.
- **Feedback** is a property of systems that means that the system has **internal loops** that **control** the system behavior.
- **Hierarchy** is a property of systems that means that the system has **levels of organization**.



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Systems Properties II

- **Equifinality** is a property of systems that means that the system can reach the **same goal** from **different paths**.
- **Permeability** is a property of systems that means that the system can **interact** with the **environment**.
- **Dissipative** is a property of systems that means that the system can **lose energy** and **information** to the environment.
- **Homeostasis** is a property of systems that means that the system can **maintain a stable state**.



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Basic Concepts

A **system** can be **classified** according to different **criteria** like **openness**, **adaptability**, **determinism**, and **linearity**.



Systems Classification I

- **Open systems** are systems that can **interact** with their **environment**.
- **Closed systems** are systems that **cannot interact** with their **environment**.
- **Adaptive systems** are systems that can **change** and **adapt** to their environment.
- **Non-adaptive systems** are systems that **cannot change** or **adapt** to their environment.



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Systems Classification II

- **Deterministic systems** are systems that produce the **same output** for the **same input**.
- **Stochastic systems** are systems that produce **different outputs** for the **same input**.
- **Linear systems** are systems that produce **proportional outputs** for **proportional inputs**.
- **Non-linear systems** are systems that produce **non-proportional outputs** for any given inputs.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

